

## Fall 2025 - Assignment 2 Detailed Development Document

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# 1. Design Document

The following is the detailed development document to provide the details steps that is not given in the previous design document.

## 1.1. Developing Environment

The following is the development environment used in this project:

- Python 3.13.9
- CrewAI v1.3.0
- MongoDB Community 8.2.1
- Docker Desktop 4.49.0
- LLM: gemini-2.0-flash-lite (prototype use)
- IDE: Visual Studio Code 1.105.1

## 1.2. Use case definition

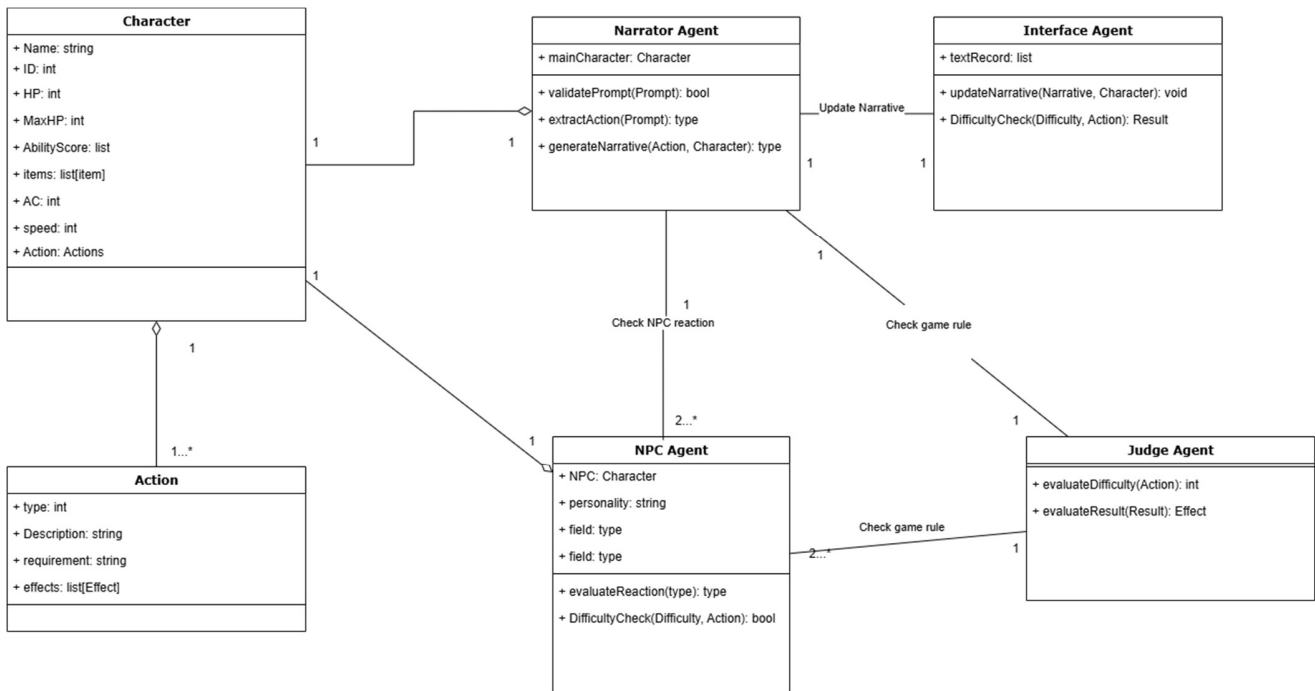
|                           |                                                                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Brief Description:</b> | User input action or information to the system to progress the story                                                                                                                    |
| <b>Precondition(s):</b>   | The game started, and the initial message was displayed                                                                                                                                 |
| <b>Post Condition(s):</b> | The system displays and updates the narrative and the story progress.                                                                                                                   |
| <b>Process Steps</b>      |                                                                                                                                                                                         |
| 1                         | The Interface agent receives the prompt from the user and passes it to Narrator agent                                                                                                   |
| 2                         | The Narrator agent looks in the prompt and identifies the action.                                                                                                                       |
| 3                         | Narrator agent gets the action difficulty from the Judge agent and passes it to the interface agent.                                                                                    |
| 4                         | The interface agent displays the difficulty and is pending for player's decision. If the user continues the action, interface agent does a random roll to determine the success or not. |
| 5                         | Judge agent evaluates and executes the consequence/effect based on the action result and informs the Narrator agent and NPC agents.                                                     |
| 6                         | NPC agents evaluate corresponding reaction and check its difficulty from the judge                                                                                                      |
| 7                         | NPC agents do a random roll to determine success or not, and inform the judge agent of the result.                                                                                      |
| 8                         | Judge agent evaluates and executes the consequence/effect based on the action result and informs the Narrator agent.                                                                    |

|                   |                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9                 | Narrator agent generates the user’s action and the NPC agent’s reaction narrative.                                                                                                                                                                                                                                                                  |                                                                                                                                                           |
| 10                | Narrator agent updates the game state according to both NPC and user narrative, if needed                                                                                                                                                                                                                                                           |                                                                                                                                                           |
| 11                | Narrator agent passes NPC agent’s reaction narrative and user action narrative to the interface agent                                                                                                                                                                                                                                               |                                                                                                                                                           |
| 12                | The interface agent shows the narratives to the user and progresses the story.                                                                                                                                                                                                                                                                      |                                                                                                                                                           |
| Exceptions        |                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                           |
| 2a                | The user’s prompt consists of invalid, conflicting, or ambiguous information or action                                                                                                                                                                                                                                                              | Narrator agent replies to the interface agent to clarify the prompt                                                                                       |
| 3a                | The judge agent determines that the action does not need a difficulty check (it is impossible to succeed, or it is too easy)                                                                                                                                                                                                                        | Judge agent informs the narrator agent and NPC agents about the effect of the action (Jump to 6) and skips the difficulty check in the interface agent.   |
| 6a                | Multiple NPC agents try to react to the user action                                                                                                                                                                                                                                                                                                 | Based on the NPC information to determine the order of execution. Append the previous player and NPC action to judge agent for evaluating the difficulty. |
| Relationship      |                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                           |
| Initiator         | User                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                           |
| Collaborator      | Interface agent, Narrator agent, Jude agent, NPC agents                                                                                                                                                                                                                                                                                             |                                                                                                                                                           |
| Data requirements |                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                           |
| User prompt       | A text input by the user, which describes the action of the main character.                                                                                                                                                                                                                                                                         |                                                                                                                                                           |
| pre-defined story | A text document defines the outline, objective, and stages of the story. Each stage consists of the environment description, NPCs, and story information.                                                                                                                                                                                           |                                                                                                                                                           |
| Rule document     | A text document defines the terminology and the game rule logic. It is based on the <a href="#">Dungeon and Dragon System Reference Document</a> (D&D SRD), which allows public use.                                                                                                                                                                |                                                                                                                                                           |
| Game State        | A state object that stores the status of the current story and facilitates the communication between tasks. It consists of stages, the main character, and a list of previous actions.<br>In stages, it contains the changes to NPCs' status and environment.<br>In main character contains the story progress based on the outline and the status. |                                                                                                                                                           |
| Game Data         | A database contains the value, effect, description of the game object, including enemy, item, class, ability.                                                                                                                                                                                                                                       |                                                                                                                                                           |

This is based on the <https://github.com/5e-bits/5e-database>. The data are all within the DnD SRD which allows public use.

### 1.3. Detailed class diagram

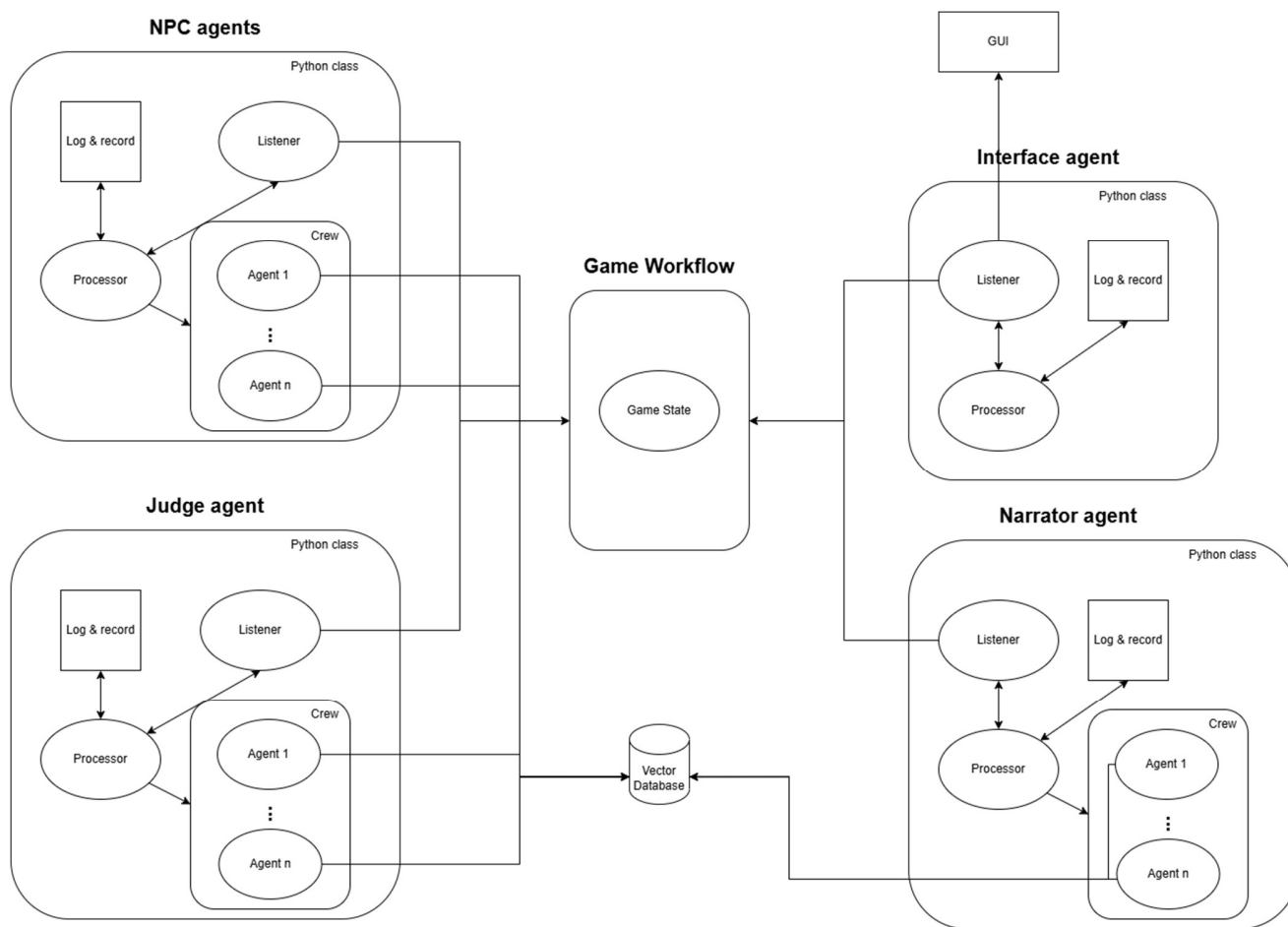
Here is detailed class diagram:



### 1.4. Project Structure

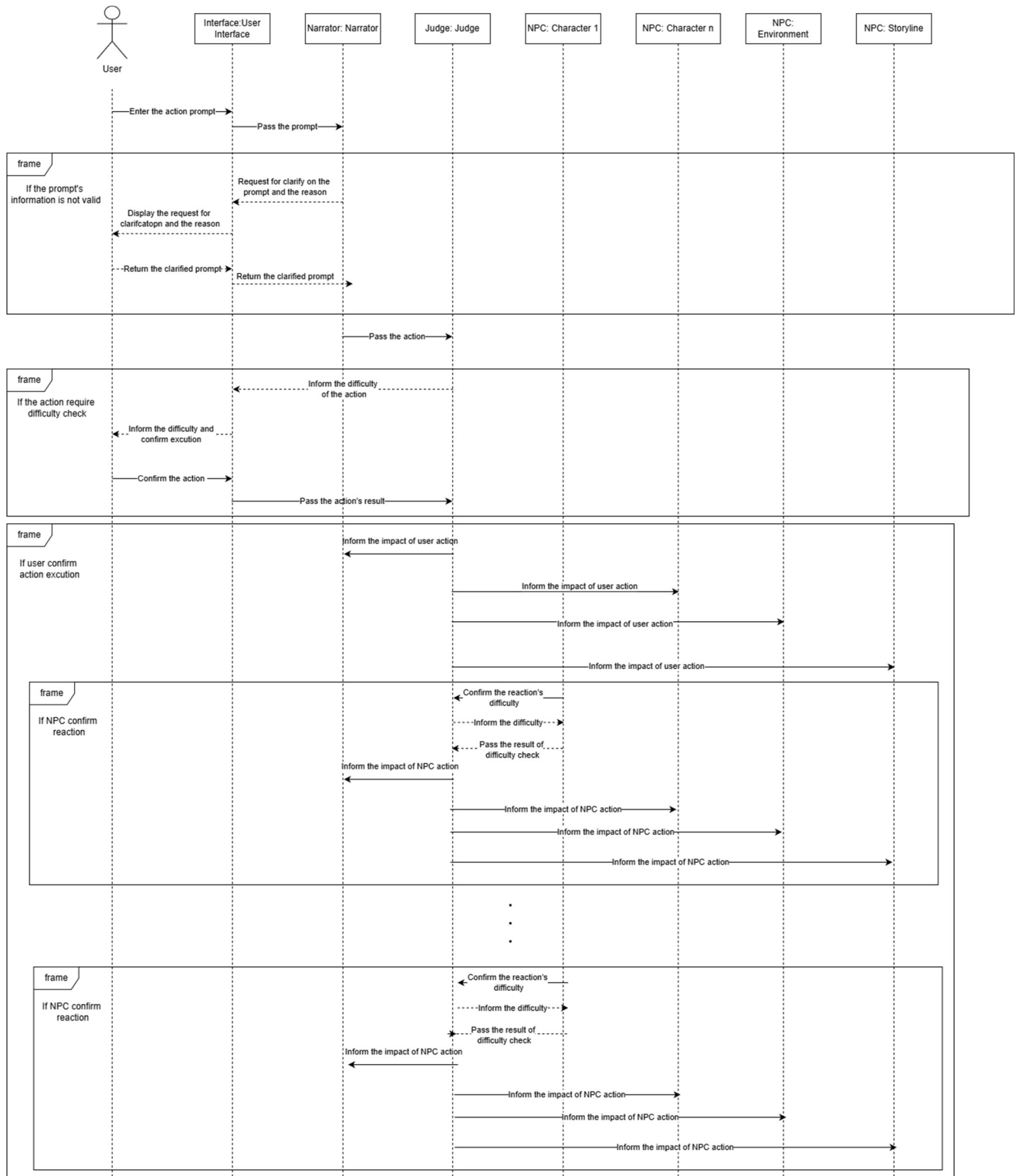
The project flow mechanic will utilize CrewAI workflow, with state-based and event-driven control.

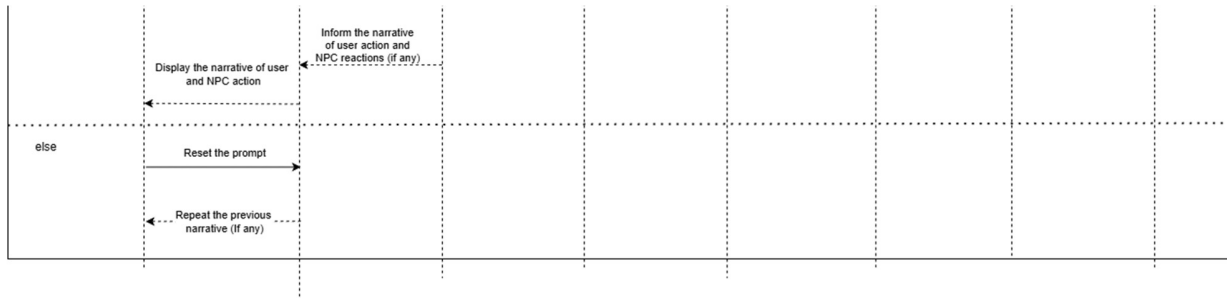
Each agent is a Python class, consisting of listener functions, processor functions, and a CrewAI crew, which includes one or more CrewAI agents.



## 1.5. Message sequence chart

Here is the revised sequence chart:





As CrewAI workflow emphasizes the state-based control, most key information is passed between the agent with the state, with some others via the database or direct message.

| Name                     | Type                      | Description                                                                                                                                 |
|--------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>current_step</b>      | str                       | Current step in the workflow,                                                                                                               |
| <b>current_agent</b>     | str                       | Name of the agent currently processing                                                                                                      |
| <b>error_log</b>         | List[Dict[str, str]]      | List of errors log                                                                                                                          |
| <b>retry_count</b>       | Dict[str, int]            | Number of retries per crew: {"crew_a": 2}                                                                                                   |
| <b>pipeline_branch</b>   | Dict[str, str]            | Current conditional branch (e.g., "success", "failed", "retry") for router logic                                                            |
| <b>start_time</b>        | str                       | Timestamp of workflow start                                                                                                                 |
| <b>last_updated</b>      | str                       | Timestamp of last state update                                                                                                              |
| <b>prompt_text</b>       | str                       | Current user's prompt being processed                                                                                                       |
| <b>Action_difficulty</b> | Dict[str, int]            | Extracted action obtained from the prompt by the Narrator Agent, and its corresponding difficulty by the Judge Agent                        |
| <b>difficulty_check</b>  | int                       | The difficulty check result obtained by the Interface agent                                                                                 |
| <b>effect</b>            | str                       | The effect of action that evaluated by Judge Agent                                                                                          |
| <b>reactions</b>         | list[Dict[NPC, str, int]] | List of reactions obtained from the NPC agents, and their corresponding difficulty by the Judge Agent                                       |
| <b>final_output</b>      | str                       | Final generated narrative of both user's action, it's effects, and NPCs reactions                                                           |
| <b>Impact_stage</b>      | List[str, str]            | The accumulated impact on the story, including the death of the NPC, triggered trap, and destroyed some buildings..... Separated in stages. |

## 1.6. Data specification

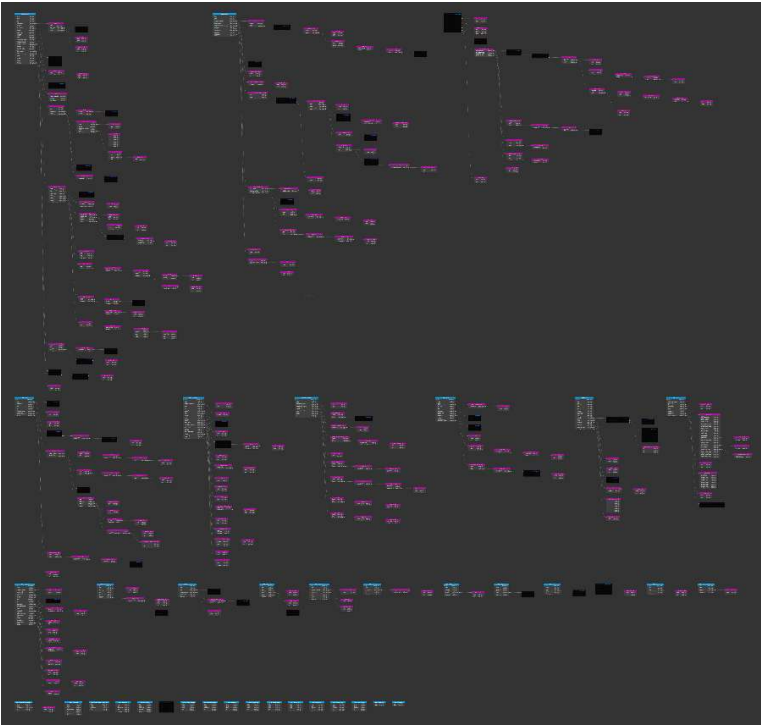
Most data are stored in local MongoDB Community 8.2, and running on docker.

Most data are retrieved in <https://github.com/5e-bits/5e-database>, here is the list of collections:

|                           |                                |                                |
|---------------------------|--------------------------------|--------------------------------|
| 2014-ability-scores       | 2014-magic-schools             | 2024-ability-scores            |
| 2014-alignments           | 2014-monsters                  | 2024-alignments                |
| 2014-backgrounds          | 2014-proficiencies             | 2024-collections               |
| 2014-classes              | 2014-races                     | 2024-conditions                |
| 2014-collections          | 2014-rule-sections             | 2024-damage-types              |
| 2014-conditions           | 2014-rules                     | 2024-equipment                 |
| 2014-damage-types         | 2014-skills                    | 2024-equipment-categories      |
| 2014-equipment            | 2014-spells                    | 2024-languages                 |
| 2014-equipment-categories | 2014-subclasses                | 2024-magic-schools             |
| 2014-feats                | 2014-subraces                  | 2024-skills                    |
| 2014-features             | 2014-traits                    | 2024-weapon-mastery-properties |
| 2014-languages            | 2014-weapon-properties         | 2024-weapon-properties         |
| 2014-levels               | 2024-weapon-mastery-properties |                                |
| 2014-magic-items          | 2024-weapon-properties         |                                |

Strings data that needed to be retrieve by LLM are generated the embeddings and vector search index.

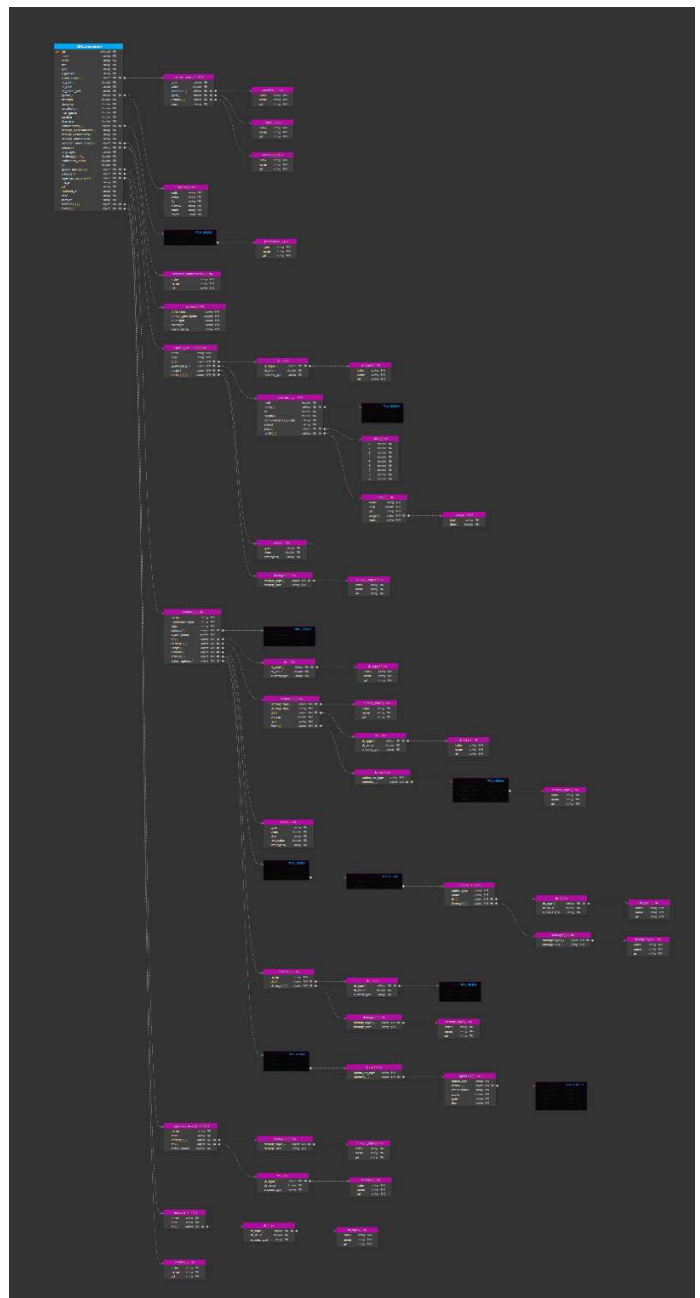
The diagram is generated by Moon Modeler 11.0.0, reverse engine from local MonogoDB



We are only using the 2014 collections and some of the fields, and here introduce some significant collections:

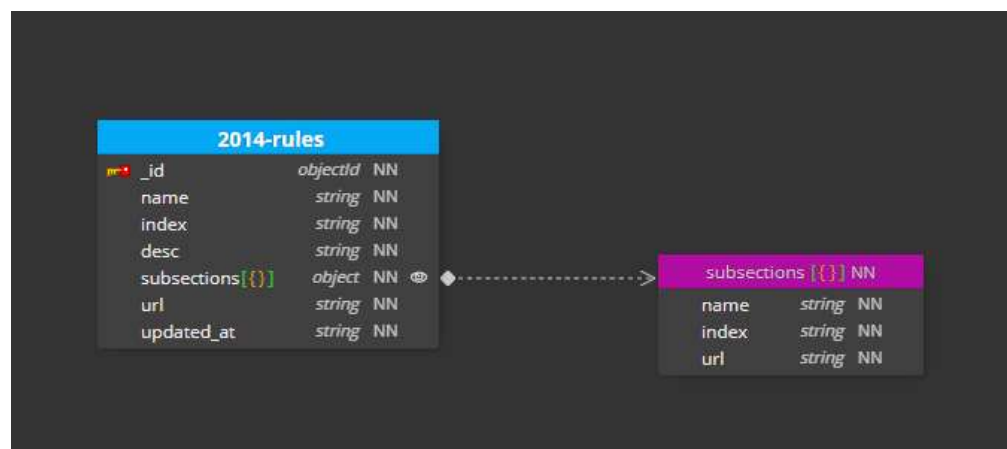
### 1.6.1. 2014-monsters

| 2014-monsters                                                                         |          | ...                                                                                                                                                                        |
|---------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  _id | objectId | NN                                                                                                                                                                         |
| index                                                                                 | string   | NN                                                                                                                                                                         |
| name                                                                                  | string   | NN                                                                                                                                                                         |
| size                                                                                  | string   | NN                                                                                                                                                                         |
| type                                                                                  | string   | NN                                                                                                                                                                         |
| alignment                                                                             | string   | NN                                                                                                                                                                         |
| armor_class[{}]                                                                       | object   | NN       |
| hit_points                                                                            | double   | NN                                                                                                                                                                         |
| hit_dice                                                                              | string   | NN                                                                                                                                                                         |
| hit_points_roll                                                                       | string   | NN                                                                                                                                                                         |
| speed {}                                                                              | object   | NN       |
| strength                                                                              | double   | NN                                                                                                                                                                         |
| dexterity                                                                             | double   | NN                                                                                                                                                                         |
| constitution                                                                          | double   | NN                                                                                                                                                                         |
| intelligence                                                                          | double   | NN                                                                                                                                                                         |
| wisdom                                                                                | double   | NN                                                                                                                                                                         |
| charisma                                                                              | double   | NN                                                                                                                                                                         |
| proficiencies[{}]                                                                     | object   | NN   |
| damage_vulnerabilities[]                                                              | string   | NN                                                                                                                                                                         |
| damage_resistances[]                                                                  | string   | NN                                                                                                                                                                         |
| damage_immunities[]                                                                   | string   | NN                                                                                                                                                                         |
| condition_immunities[{}]                                                              | object   | NN   |
| senses {}                                                                             | object   | NN   |
| languages                                                                             | string   | NN                                                                                                                                                                         |
| challenge_rating                                                                      | double   | NN                                                                                                                                                                         |
| proficiency_bonus                                                                     | double   | NN                                                                                                                                                                         |
| xp                                                                                    | double   | NN                                                                                                                                                                         |
| special_abilities[{}]                                                                 | object   | NN   |
| actions[{}]                                                                           | object   | NN   |
| legendary_actions[{}]                                                                 | object   | NN   |
| image                                                                                 | string   | NN                                                                                                                                                                         |
| url                                                                                   | string   | NN                                                                                                                                                                         |
| updated_at                                                                            | string   | NN                                                                                                                                                                         |
| desc                                                                                  | string   | NN                                                                                                                                                                         |
| subtype                                                                               | string   | NN                                                                                                                                                                         |
| reactions[{}]                                                                         | object   | NN   |
| forms[{}]                                                                             | object   | NN   |

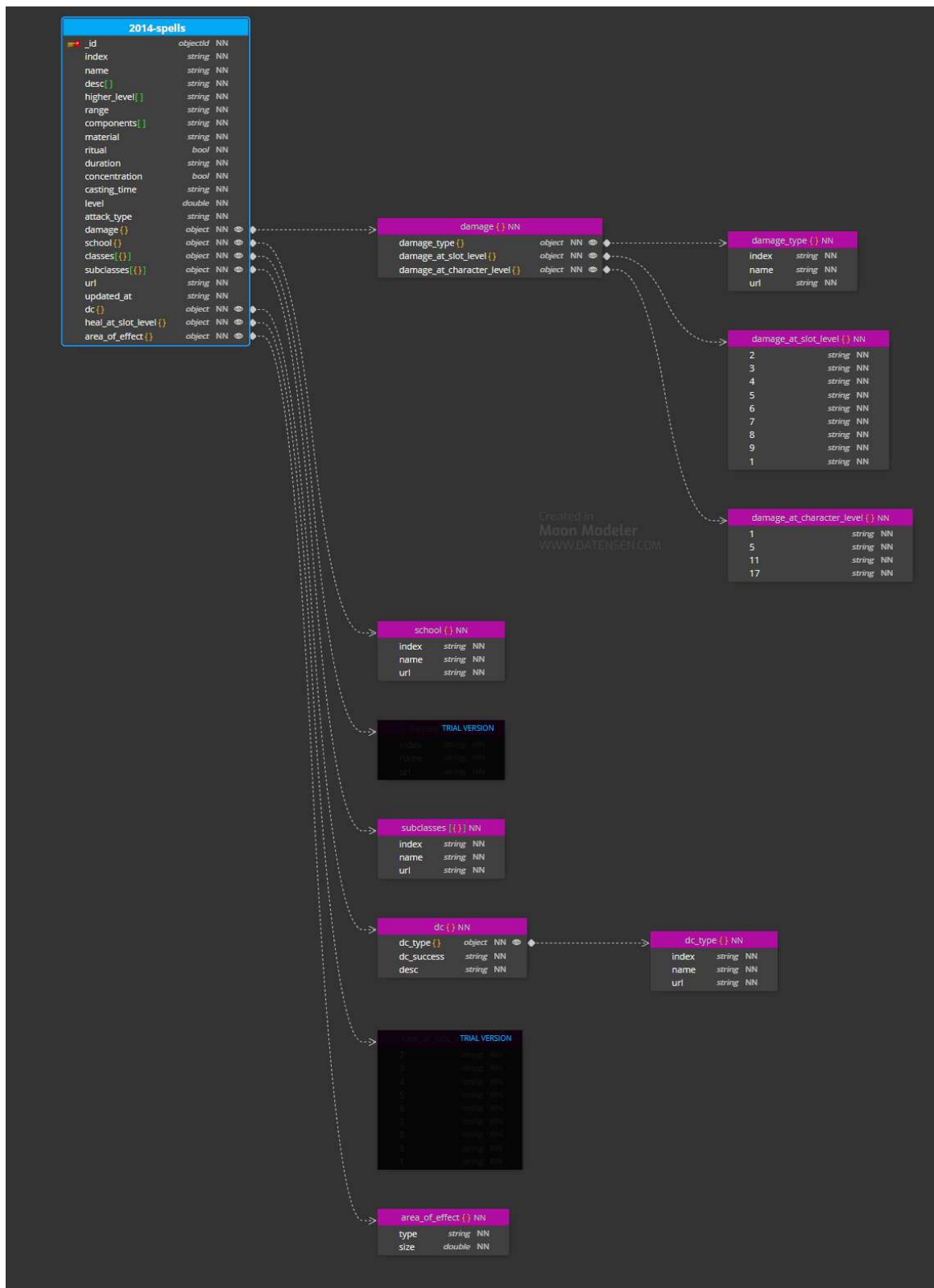




### 1.6.2. 2014-rules



### 1.6.3. 2014-spells



#### 1.6.4. Other collections

Some data is not coming from the 5e-database, specifically the pre-defined story. Here is the structure of them:

| Table Name        | Attributes | Type     | Description                                                                        |
|-------------------|------------|----------|------------------------------------------------------------------------------------|
| pre-defined story | _id        | ObjectId | ID                                                                                 |
|                   | name       | string   | Name of the story                                                                  |
|                   | stages[]   | object   | List of stages in the storyline.                                                   |
|                   | objective  | string   | The objective defines the end criteria of the story                                |
|                   | outline    | string   | The default progression of the story                                               |
|                   | mapDesc    | string   | The description defines the map which the story takes place, including all stages. |

| Table Name | Attributes     | Type     | Description                                                                                                       |
|------------|----------------|----------|-------------------------------------------------------------------------------------------------------------------|
| Stage      | _id            | ObjectId | ID                                                                                                                |
|            | name           | string   | Name of the Stage                                                                                                 |
|            | envDesc        | string   | Description of the environment element, including visual information, the layout of stage.....                    |
|            | venues[]       | string   | List of venues in this stage, including room, building, sections.....                                             |
|            | storyDesc      | string   | Description of the story-related information, including hints to progress the story, and background of the stage. |
|            | startVenue     | Object   | The start venue that the story begin.                                                                             |
|            | startnarrative | string   | The narrative shown at the start, before the user enters his first prompt.                                        |

| Table Name | Attributes      | Type     | Description                                                                                                                         |
|------------|-----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------|
| Venue      | _id             | ObjectId | ID                                                                                                                                  |
|            | name            | string   | Name of the venue                                                                                                                   |
|            | envDesc         | string   | Description of the environment element, including visual information, the layout of the venue, interactive elements like traps..... |
|            | NPCs[]          | Object   | List of NPCs exists in the current venues.                                                                                          |
|            | connectVenues[] | Object   | List of connected venues. Some interaction may involve these venues, specifically their NPCs.                                       |
|            | storyDesc       | string   | Description of the story-related information, including                                                                             |
|            | action[]        | Object   | List of supported actions in this stage, like investigation, move to other stages or venues, communicate with NPC                   |

| Table Name | Attributes | Type | Description |
|------------|------------|------|-------------|
|------------|------------|------|-------------|

|     |             |               |                                                                                                     |
|-----|-------------|---------------|-----------------------------------------------------------------------------------------------------|
| NPC | _id         | ObjectId      | ID                                                                                                  |
|     | name        | string        | Name of the NPC                                                                                     |
|     | Desc        | string        | Description of the NPC                                                                              |
|     | stat        | 2014-monsters | All status and information define the NPC combat ability, including hit point, combat actions.....  |
|     | action[]    | Object        | List of story-related interactions, including persuading, frightening, investigating, stealing..... |
|     | Intention   | string        | Define the short-term intention that was used to evaluate the NPC's reaction                        |
|     | target      | string        | Define the long-term target on the story that was used to evaluate the NPC's intention              |
|     | personality | string        | A constant string to define the personality of the NPC, used to evaluate the intention              |

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